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CHEMICAL REACTION ENGINEERING

FINAL EXAM

CHE 321, SPRING 2025, CLOSED BOOK, TIME: 60 MIN

The endothermic gas phase reaction $A \rightarrow B + C$ is carried out in a single steam-jacketed fluidized bed reactor, where the temperature of the steam circulating through the jacket remains constant at $T_a = 300^\circ\text{C}$.

For the following data, calculate the steady-state reactor temperature.

Conversion: $X = 85\%$

Heat-transfer area of the jacket: $A = 5 \text{ m}^2$

Overall heat transfer coefficient of the jacket: $U = 500 \frac{\text{J}}{\text{s.m}^2.\text{K}}$

Heat of reaction per unit mole of species A at 25°C : $\Delta_r H = 120 \frac{\text{kJ}}{\text{mol}} \rightarrow 120,000 \frac{\text{J}}{\text{mol}}$

Feed temperature: $T_0 = 40^\circ\text{C} = 313\text{K}$

Feed molar flow rates: $F_{A0} = 10 \frac{\text{mol}}{\text{s}}$, $F_{I0} = 2 \frac{\text{mol}}{\text{s}}$ (inert), $F_{B0} = F_{C0} = 0$

Heat capacities (assume constant) $c_{pA} = 90 \frac{\text{J}}{\text{mol.K}}$, $c_{pB} = 60 \frac{\text{J}}{\text{mol.K}}$, $c_{pC} = 40 \frac{\text{J}}{\text{mol.K}}$, $c_{pI} = 50 \frac{\text{J}}{\text{mol.K}}$

Reactor energy balance: $\frac{dU}{dt} = \sum_{\alpha} F_{\alpha 0} h_{\alpha}(T_0) - \sum_{\alpha} F_{\alpha} h_{\alpha}(T) + \dot{Q} + \dot{W}_s$

Conversion $X \equiv \frac{F_{A0} - F_A}{F_{A0}}$

FBR

$$T_a = 573\text{K}$$

$$T_R = 298$$

$$\theta_I = \frac{2}{10} = \frac{1}{5} = 0.2$$

Energy Balance

$$0 = \sum_{\alpha} F_{\alpha 0} h_{\alpha}(T_0) - \sum_{\alpha} F_{\alpha} h_{\alpha}(T) + \dot{Q} \quad F_{\alpha} = F_{\alpha 0} + Y_{\alpha} F_{A0} X$$

$$\rightarrow U(T_a - T)A$$

$$\Rightarrow \sum_{\alpha} F_{\alpha 0} h_{\alpha}(T_0) - \sum_{\alpha} F_{\alpha 0} h_{\alpha}(T) - F_{A0} X \sum_{\alpha} Y_{\alpha} h_{\alpha}(T) + \dot{Q} = 0$$

$$\Rightarrow \sum_{\alpha} F_{\alpha 0} (h_{\alpha}(T_0) - h_{\alpha}(T)) - F_{A0} X \Delta H_r(T) + \dot{Q} \rightarrow \sum_{\alpha} F_{\alpha 0} c_{p\alpha} (T_0 - T) - F_{A0} X (\Delta H_r(T_R) + \Delta C_{p,r} (T - T_R))$$

$$\Rightarrow \sum_{\alpha} F_{\alpha 0} c_{p\alpha} (T_0 - T) - F_{A0} X (\Delta H_r(T_R) + \Delta C_{p,r} (T - T_R)) + \dot{Q} = 0 \quad \Delta C_{p,r} = \sum_{\alpha} Y_{\alpha} c_{p\alpha}$$

$$\Rightarrow (T_0 - T) \sum_{\alpha} F_{\alpha 0} c_{p\alpha} - F_{A0} X \Delta H_r(T_R) - F_{A0} X \Delta C_{p,r} (T - T_R) + U(T_a - T)A = 0$$

$$\Rightarrow T \left(\sum_{\alpha} F_{\alpha 0} c_{p\alpha} \right) + T (F_{A0} X \Delta C_{p,r}) + T(UA) = F_{A0} X (T_R \Delta C_{p,r} - \Delta H_r(T_R)) + T_0 \sum_{\alpha} F_{\alpha 0} c_{p\alpha} + T_a UA$$

$$\Rightarrow T = \frac{F_{A0} X (T_R \Delta C_{p,r} - \Delta H_r(T_R)) + T_0 \sum_{\alpha} F_{\alpha 0} c_{p\alpha} + T_a UA}{\sum_{\alpha} F_{\alpha 0} c_{p\alpha} + F_{A0} X \Delta C_{p,r} + UA}$$

Name: $F_{A0}X = 10(0.85) = 8.5 \frac{\text{mol}}{\text{s}}$, $T_R = 298\text{K}$, $T_a = 573\text{K}$, $T_o = 313\text{K}$

$$\Delta C_{pr} = \sum_{\alpha} Y_{\alpha} C_{p\alpha} = (-1)(90) + (1)(60) + (1)(40) = 10 \frac{\text{J}}{\text{mol} \cdot \text{K}}, \Delta H_a(T_R) = 120,000 \frac{\text{J}}{\text{mol}}$$

$$\sum_{\alpha} F_{A0} C_{p\alpha} = (10)(90) + (2)(50) = 1000 \frac{\text{J}}{\text{s} \cdot \text{K}}$$

$$T_a U A = (573\text{K}) \left(500 \frac{\text{J}}{\text{m}^2 \cdot \text{K}} \right) (5\text{m}^2) = 1432500 \frac{\text{J}}{\text{s}}, U A = (500)(5) = 2500 \frac{\text{J}}{\text{K} \cdot \text{s}}$$

$$\Rightarrow T = \frac{F_{A0}X [T_R \Delta C_{pr} - \Delta H_a(T_R)] + T_o \sum_{\alpha} F_{A0} C_{p\alpha} + T_a U A}{\sum_{\alpha} F_{A0} C_{p\alpha} + F_{A0}X \Delta C_{pr} + U A}$$

$$\Rightarrow T = \frac{(8.5 \frac{\text{mol}}{\text{s}}) [(298\text{K})(10 \frac{\text{J}}{\text{mol} \cdot \text{K}}) - 120,000 \frac{\text{J}}{\text{mol}}] + (313\text{K})(1000 \frac{\text{J}}{\text{s} \cdot \text{K}}) + 1432500 \frac{\text{J}}{\text{s}}}{(1000 \frac{\text{J}}{\text{s} \cdot \text{K}}) + (8.5 \frac{\text{mol}}{\text{s}})(10 \frac{\text{J}}{\text{mol} \cdot \text{K}}) + 2500 \frac{\text{J}}{\text{K} \cdot \text{s}}}$$

$$\Rightarrow T = 209.437\text{K}$$

$$\frac{\frac{\text{J}}{\text{s}}}{\frac{\text{J}}{\text{K} \cdot \text{s}}} \rightarrow \frac{1}{1/\text{K}} = \text{K}$$

